

RetailBusiness360

Microsoft Fabric Data Engineering Platform

End-to-End Retail Analytics | Medallion Architecture | Real-Time Streaming

1. Executive Summary

RetailBusiness360 is a production-grade, end-to-end retail analytics platform built entirely on Microsoft Fabric. The platform unifies four distinct data source systems — an on-premises ERP, a CRM system, a logistics REST API, and a real-time point-of-sale event stream — into a single, governed Medallion Lakehouse architecture that progressively refines raw data through Bronze, Silver, and Gold layers.

The solution processes over 172,000 records spanning 30 international retail stores across the UK, USA, Germany, India, Australia, and UAE. It delivers analytics-ready dimensional models and aggregated Gold tables powering business intelligence dashboards for sales performance, customer behaviour, inventory health, and logistics KPIs.

Dimension	Value
Total Data Records	172,036 rows across 8 source datasets
Stores Covered	30 stores across 6 countries
Data Sources	4 (ERP on-prem, CRM, REST API, Event Stream)
Ingestion Pipelines	4 (daily, weekly, streaming)
Transformation Notebooks	2 (Silver & Gold layer PySpark)
Gold Aggregations Produced	6 analytical tables
Tech Stack	Microsoft Fabric, PySpark, Delta Lake, KQL, Azure Event Hub

2. Business Objective

RetailBusiness360 was designed to solve a common retail analytics challenge: data is generated across multiple siloed systems with different formats, schedules, and protocols. The platform's objective is to consolidate all operational data into a unified, continuously refreshed Lakehouse that enables:

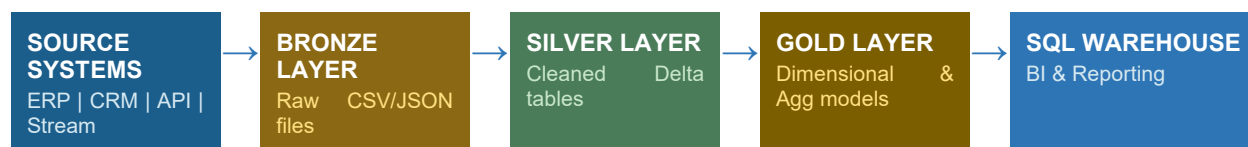
- Real-time and near-real-time transaction monitoring across all sales channels
- 360-degree customer profiling combining purchase history, CRM interactions, and loyalty tier
- Supply chain and logistics performance tracking by delivery partner and shipment route
- Inventory health alerts and reorder recommendations by warehouse and product
- Multi-dimensional sales reporting broken down by product, region, day, and customer segment

The platform is designed for use by retail operations teams, supply chain managers, and data analysts who require reliable, daily-refreshed data with governed quality standards and traceable lineage from source to Gold.

3. Architecture Overview

The platform implements the Medallion Architecture pattern — a staged data quality design where data quality, richness, and analytical readiness increase progressively across three lake layers. All storage is hosted on Microsoft Fabric's OneLake via a single Lakehouse artifact.

3.1 Architecture Diagram (Logical)



3.2 Layer Descriptions

Bronze Layer — Raw Ingestion

The Bronze layer stores all data exactly as received from source systems. Files are landed under structured folder paths (e.g. Bronze/ERP/inventory/, Bronze/CRM/customers/) with date-stamped filenames for ERP files. No transformation is applied; Bronze is an immutable audit trail. Pipelines write CSV, JSON, or Delta format depending on source type.

Silver Layer — Cleaned & Standardised

The Silver layer applies PySpark transformations to produce clean, typed, deduplicated datasets written as Delta Lake tables. Quality rules enforced at this layer include: type casting for all numerics and dates, null handling and default fills, string normalisation (trim + upper), negative value filtering for prices and quantities, and deduplication on primary keys. An audit column (`_silver_load_ts`) is added to every table for lineage tracking.

Gold Layer — Business Aggregations

The Gold layer joins Silver tables and produces dimensional models and pre-aggregated analytical tables optimised for BI consumption. Six Gold tables are produced covering sales, products, customers, regions, inventory, and shipment KPIs. A derived RFM segmentation (Champion / Loyal / Potential / New) is computed per customer. Revenue rank is computed per product. Reorder flags are derived for inventory.

4. Data Sources

The platform ingests data from four source system types, each representing a distinct integration pattern in enterprise retail environments.

Source System	Tables / Files	Records	Format	Ingestion Frequency
ERP (On-Premises SQL Server)	Products, Inventory	2,000 products / 50,000 inventory	CSV (date-stamped)	Daily at 6 PM
CRM (Cloud / Lakehouse)	Customers, Customer Interactions	10,000 customers / 10,000 interactions	Delta (partitioned)	Daily at 6 AM
Logistics REST API	Shipments	25,000 shipments	JSON (REST GET)	Weekly
POS Event Stream (Azure Event Hub)	Sales Transactions	100,000 transactions	Streaming events via KQL	Real-time / continuous
POS Static (Lakehouse)	Stores	30 stores	CSV	On-demand / manual
Social Media	Product Reviews	~1,918 KB text	Text file (REVIEW)	Batch periodic

4.1 Dataset Schemas

Customers

Field	Type	Description
customer_id	String (PK)	Unique customer identifier (e.g. C000001)
full_name	String	Customer full name
gender	String	Gender (normalised to upper case)
age	Integer	Customer age
city	String	City of residence
country	String	Country of residence
signup_date	Date	Account creation date
loyalty_tier	String	Bronze / Silver / Gold / Platinum (default: Bronze)

Sales Transactions

Field	Type	Description
transaction_id	String (PK)	Unique transaction identifier
customer_id	String (FK)	Linked customer
product_id	String (FK)	Linked product
quantity	Integer	Units purchased (must be > 0)
sales_amount	Double	Revenue value in local currency (must be > 0)
payment_method	String	Card / Cash / Online / Wallet etc.
sales_channel	String	In-Store / Online / Mobile
store_id	String (FK)	Linked store
transaction_timestamp	Timestamp	Date and time of transaction

Shipments

Field	Type	Description
shipment_id	String (PK)	Unique shipment identifier
order_id	String	Linked order reference
customer_id	String (FK)	Linked customer
product_id	String (FK)	Linked product
status	String	Shipped / Delivered / In Transit / Returned
delivery_partner	String	Courier company name
origin_city / destination_city	String	Route endpoints
ship_date / delivery_date	Date	Logistics dates
delivery_days	Integer	Actual days to deliver (recalculated if null)
weight_kg / shipping_cost	Double	Package weight and shipping fee

5. Ingestion Pipelines

Four Microsoft Fabric Data Factory pipelines handle source-to-Bronze ingestion. Each pipeline is defined as an ARM-style JSON template and parameterised for workspace and artifact IDs, enabling deployment to any Fabric workspace.

5.1 ERP_Daily_6PM_OnPrem2Bronze

Ingests all base tables from the on-premises SQL Server database (retail360) into the Lakehouse Bronze zone.

Step	Activity Type	Details
1	Lookup	Queries INFORMATION_SCHEMA.TABLES to dynamically discover all tables — no hardcoded table list
2	ForEach (sequential)	Iterates over each discovered table name
3 (inner)	Copy Activity	Reads from SqlServerSource, writes to Lakehouse Files under Bronze/ERP/<TableName>/<TableName>_yyyyMMdd.csv

- Source connection: on-premises SQL Server via self-hosted Integration Runtime (falcon;retail360)
- Sink format: CSV with headers, comma-delimited, date-stamped filenames
- Scheduled: daily at 6 PM, 12-hour timeout per activity

5.2 CRM_Daily_6AM_Lakehouse2SQLWarehouse

Copies CRM data (customers and interactions) from the Lakehouse Files zone (CRM_RAW folder) into the Fabric SQL Warehouse for reporting.

Step	Activity Type	Details
1	Set Variable	Sets TableList = ['customers', 'interactions']
2	ForEach (sequential)	Iterates over the two table names
3 (inner)	Copy Activity	Reads DelimitedText from CRM_RAW/<table>.csv, inserts into org.<table> in SQL Warehouse
4	RefreshDataflow	Triggers a Fabric Dataflow Gen2 refresh (dataflowId: a489c998) after load completes

- Scheduled: daily at 6 AM
- Sink: Fabric SQL Database with insert write behaviour, schema org

5.3 Logistics_Weekly_API

Pulls shipment data from a Logistics REST API and lands it in the Lakehouse Bronze zone.

Step	Activity Type	Details
1	Copy Activity (REST source)	HTTP GET to shipments.json endpoint with RFC5988 pagination support
Mapping	Column mapping	Explicit 13-field mapping: shipment_id, order_id, customer_id, product_id, status, delivery_partner, origin/destination city, ship/delivery dates, delivery_days, weight_kg, shipping_cost
Sink	Lakehouse CSV	Written to Bronze/LOGISTICS/shipments/shipments.csv

- Frequency: weekly
- Nested JSON is flattened using columnFlattenSettings with dot-notation delimiter

5.4 Streaming_KQL_2_SQL

Bridges real-time POS transaction data from the Fabric EventHouse (KQL database) into the Fabric SQL Warehouse for immediate analytical use.

Step	Activity Type	Details
1	Copy Activity (Kusto source)	Reads from KQL table RBusiness360 at endpoint trd-b7b61ns6peacx0kd1s.z4.kusto.fabric.microsoft.com
Mapping	Column mapping	9-field explicit mapping: transaction_id, customer_id, product_id, quantity, sales_amount, payment_method, sales_channel, store_id, transaction_timestamp
Sink	FabricSqlDatabase	Auto-creates org.transactions table if absent, insert mode

- Event stream is triggered by EventStream_Trigger.ipynb, which uses azure-eventhub SDK to stream CSV rows as JSON events at 1-second intervals
- KQL database serves as the buffer between the event stream and the SQL sink

6. Transformation Notebooks

6.1 Transformation_1 — Bronze to Silver (Data Cleaning)

This PySpark notebook reads raw Bronze files and produces clean, typed Delta tables in the Silver zone. All tables include a `_silver_load_ts` audit timestamp.

Table	Key Transformations Applied
Customers	Cast age to Integer; parse signup_date; trim/upper gender; fillna loyalty_tier='Bronze'; dropna customer_id; dedup customer_id
Interactions	Cast interaction_id to Integer; parse interaction_date; trim interaction_type/agent_name; dropna on interaction_id and customer_id
Stores	Trim store_name, city, region; dropna store_id; dedup store_id
Products	Cast unit_price to Double, stock_quantity to Integer; filter price > 0 and quantity >= 0; dropna product_id; dedup
Inventory	Cast stock_received/dispatched to Integer; parse inventory_date (7-decimal precision); derive net_stock = received - dispatched
Transactions	Cast quantity/sales_amount; parse transaction_timestamp; derive transaction_date; filter amount > 0 and quantity > 0; dedup
Shipments	Parse ship_date and delivery_date; cast weight_kg, shipping_cost, delivery_days; recalculate delivery_days via datediff where null; dedup

6.2 Transformation_2 — Silver to Gold (Business Aggregations)

This PySpark notebook reads Silver tables and produces dimensional models and aggregated analytical datasets in the Gold zone (Parquet format).

Dimensional Tables

Table	Source Tables	Key Additions
dim_customers	Customers + Interactions	total_interactions, last_interaction_date, complaint_count (left joined from interaction summary)
dim_products	Products	Clean passthrough with <code>_gold_load_ts</code>
dim_stores	Stores	Clean passthrough with <code>_gold_load_ts</code>

Fact & Aggregation Tables

Table	Grain	Key Metrics
fact_sales	One row per transaction	Enriched with product, customer, store dims; expected_amount, discount_amount, discount_pct, year/month/quarter/week partitions
agg_sales_by_day	Per transaction_date	transaction_count, total_revenue, units_sold, avg_order_value, total_discounts
agg_sales_by_product	Per product	total_revenue, units_sold, unique_customers, avg_discount_pct, revenue_rank (Window rank)
agg_sales_by_customer	Per customer	lifetime_revenue, total_orders, first/last purchase dates, avg_order_value, distinct_products_bought, days_since_last_purchase, rfm_segment
agg_sales_by_region	Per region + year + quarter	total_revenue, units_sold, transaction_count, unique_customers
agg_inventory_health	Per warehouse + product (latest snapshot)	stock_on_hand, reorder_flag (< 10% of received), stock_value (= units x unit_price)
agg_shipment_kpis	Per delivery_partner + status	shipment_count, avg/min/max delivery_days, avg/total_shipping_cost, avg_weight_kg

Customer RFM Segmentation

Segment	Condition (lifetime_revenue)
Champion	> £5,000
Loyal	> £2,000
Potential	> £500
New	<= £500

7. Technology Stack

Category	Technology	Role in Platform
Cloud Platform	Microsoft Fabric	Unified analytics platform hosting all artifacts
Storage	OneLake (Azure Data Lake Gen2)	Single-copy storage for Bronze, Silver, and Gold layers
Lakehouse	Fabric Lakehouse (RBusiness360_LakeHouse)	Unified Files and Tables namespace for all lake data
Orchestration	Fabric Data Factory Pipelines	4 ingestion pipelines for all source systems
Batch Processing	PySpark (Fabric Spark Notebooks)	Bronze-to-Silver and Silver-to-Gold transformations
Table Format	Delta Lake (Silver) / Parquet (Gold)	ACID-compliant storage with time-travel support
Real-Time Streaming	Azure Event Hub + Fabric EventStream	POS transaction streaming pipeline
KQL Analytics	Fabric EventHouse (KQL Database)	Real-time query buffer for streaming transactions
Analytical Serving	Fabric SQL Warehouse	SQL-queryable endpoint for BI tools
Dataflow Refresh	Fabric Dataflow Gen2	CRM post-load refresh triggered by pipeline
Source (ERP)	On-Premises SQL Server (via IR)	ERP source: products, inventory, stores, transactions
Source (CRM)	Fabric Lakehouse (CRM_RAW)	CRM source: customers, interactions
Source (Logistics)	REST API (HTTP GET)	Logistics: shipments JSON endpoint
Event Trigger	Python + azure-eventhub SDK	Simulates real-time POS streaming from CSV

8. Data Quality Framework

Data quality is enforced at the Silver transformation layer through systematic PySpark rules applied to every entity before writing to Delta Lake.

Quality Rule	Implementation	Tables Applied
Type Casting	<code>.cast(IntegerType()), .cast(DoubleType())</code>	All tables
Date / Timestamp Parsing	<code>F.to_date(), F.to_timestamp()</code> with explicit format strings	Customers, Interactions, Inventory, Transactions, Shipments
String Normalisation	<code>F.trim(), F.upper()</code> on categorical string columns	All string columns
Null Key Removal	<code>.dropna(subset=[primary_key])</code>	All tables — mandatory primary key present
Default Fill	<code>.fillna({'loyalty_tier': 'Bronze'})</code>	Customers
Deduplication	<code>.dropDuplicates([primary_key])</code>	All tables on primary key
Business Rule Filtering	<code>.filter(price > 0), .filter(quantity >= 0)</code>	Products, Transactions
Derived Field Validation	<code>delivery_days</code> recalculated via <code>datediff</code> where null	Shipments
Audit Timestamps	<code>_silver_load_ts, _gold_load_ts</code> added to every table	All Silver and Gold tables
Inventory Calculation	<code>net_stock = stock_received - stock_dispatched</code>	Inventory

9. Key Engineering Outcomes

- Fully automated, parameterised ingestion pipelines covering all four source integration patterns (batch SQL, lakehouse CSV, REST API, real-time streaming)
- Scalable Medallion Architecture with clear separation of raw, cleaned, and analytical data, enabling independent evolution of each layer
- Production-grade data quality enforcement: every Silver table enforces primary key constraints, type safety, deduplication, and audit lineage
- Real-time POS data path from Azure Event Hub through KQL EventHouse to Fabric SQL Warehouse, enabling sub-minute transaction visibility
- Six analytical Gold tables covering all core retail KPIs: sales, product performance, customer RFM, regional breakdown, inventory health, and logistics SLAs
- Dynamic ERP ingestion using INFORMATION_SCHEMA discovery — no hardcoded table names, automatically picks up new tables
- International retail coverage: 30 stores across UK, USA, Germany, India, Australia, and UAE, with regional segmentation built into the data model
- Reorder alerting logic embedded in inventory Gold table: flags any product-warehouse combination with stock on hand below 10% of received stock
- Customer RFM segmentation derived automatically from transaction history — Champion, Loyal, Potential, and New tiers

11. Appendix — Store Network

The platform covers 30 stores across 6 countries, segmented into regional groupings for analytical aggregation.

Store ID	Store Name	City	Region / Country
LON001–003	London Central / East / West	London	South — UK
MAN001–002	Manchester Store / Trafford	Manchester	North — UK
BIR001–002	Birmingham Store / City	Birmingham	Midlands — UK
LEE001	Leeds Central	Leeds	North — UK
GLA001	Glasgow Store	Glasgow	Scotland — UK
EDI001	Edinburgh Store	Edinburgh	Scotland — UK
BRI001	Bristol Central	Bristol	South West — UK
LIV001	Liverpool ONE	Liverpool	North — UK
SHE001	Sheffield Store	Sheffield	North — UK
NOT001	Nottingham Central	Nottingham	Midlands — UK
CAR001	Cardiff Store	Cardiff	Wales — UK
NYC001–002	New York Fifth Ave / Brooklyn	New York	North East — USA
LAX001	Los Angeles Central	Los Angeles	West — USA
CHI001	Chicago Michigan Ave	Chicago	Midwest — USA
HOU001	Houston Galleria	Houston	South — USA
BER001	Berlin Mitte	Berlin	East — Germany
MUN001	Munich Central	Munich	South — Germany
HAM001	Hamburg Store	Hamburg	North — Germany
MUM001	Mumbai Colaba	Mumbai	West — India
DEL001	Delhi Connaught Place	Delhi	North — India
BAN001	Bangalore MG Road	Bangalore	South — India
SYD001	Sydney CBD	Sydney	NSW — Australia
MEL001	Melbourne Collins St	Melbourne	VIC — Australia

Store ID	Store Name	City	Region / Country
DXB001	Dubai Mall	Dubai	Dubai — UAE
AUH001	Abu Dhabi Central	Abu Dhabi	Abu Dhabi — UAE

End of Report — RetailBusiness360 Data Platform | May 2026